24 Cr CROMO

Every factory runs on reasons nobody remembers.

CROMO reads a plant's own archive and recovers why each process step exists - finding the waste and protecting the safeguards.



Fossil · Fence · Unknown

AMD Instinct MI300X · ROCm · vLLM · Evolus

€374,183/yr
in 12 fossil steps whose reason is gone

6 fences
protected

17 unknown
sent to a person

1975 · Primo Levi, The Periodic Table, "Chromium"

A batch of anti-rust paint keeps turning to jelly.

A young chemist traces it to one bad pigment delivery and fixes it by adding ammonium chloride. **Twenty years later** the bad pigment is long gone - and the factory still adds ammonium chloride to every batch. Nobody can say why.

The opposite also happens. Chesterton (1929): *don't tear down a fence until you know why it was put up.* Some steps that look pointless are load-bearing.

Cromo is Italian for chromium.

The problem

Recipes are full of steps whose reasons retired with the people who wrote them.

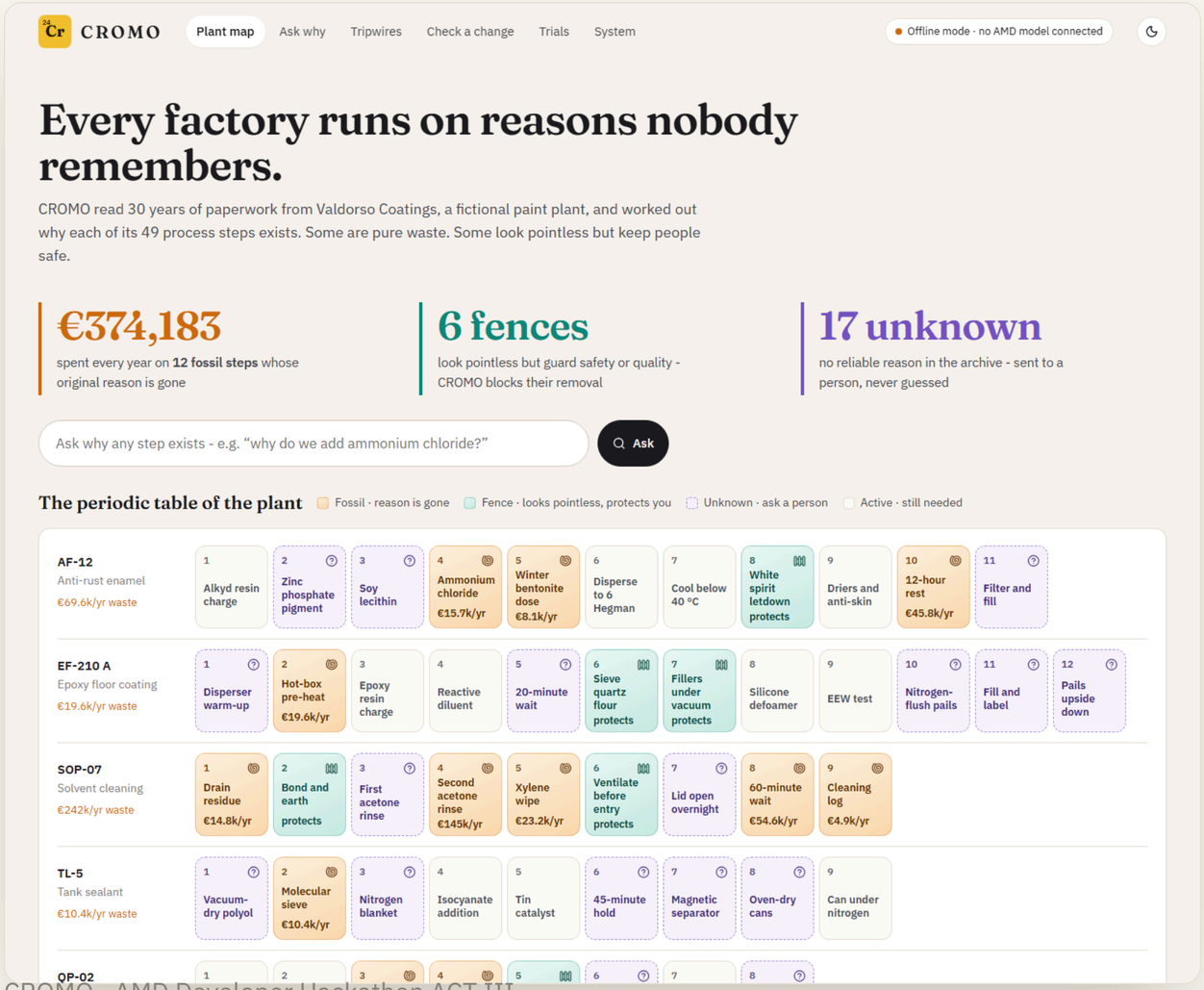
Fossils cost money every batch: material, labour, energy, tank time. Nobody dares remove them.

Fences look like waste but prevent fires, recalls and claims. Remove one and the old accident comes back.

And AI optimisers now propose "remove this, shorten that" - without knowing which is which.

Users: process engineers · quality / EHS managers · plant managers. A human approves every change.

The periodic table of a plant.



€374,183/yr

in **12 fossil steps** of a synthetic mid-sized coatings plant

6 fences

protected from removal

17 unknown

sent to a person - never guessed

Live investigation · the Levi case, recreated in our archive

It digs, it does not chat.

Recipe step first appears 1996

1995 deviation: enamel turns to gel

1995 lab test: pigment lot too alkaline

1995 change request: add NH4Cl "temporarily"

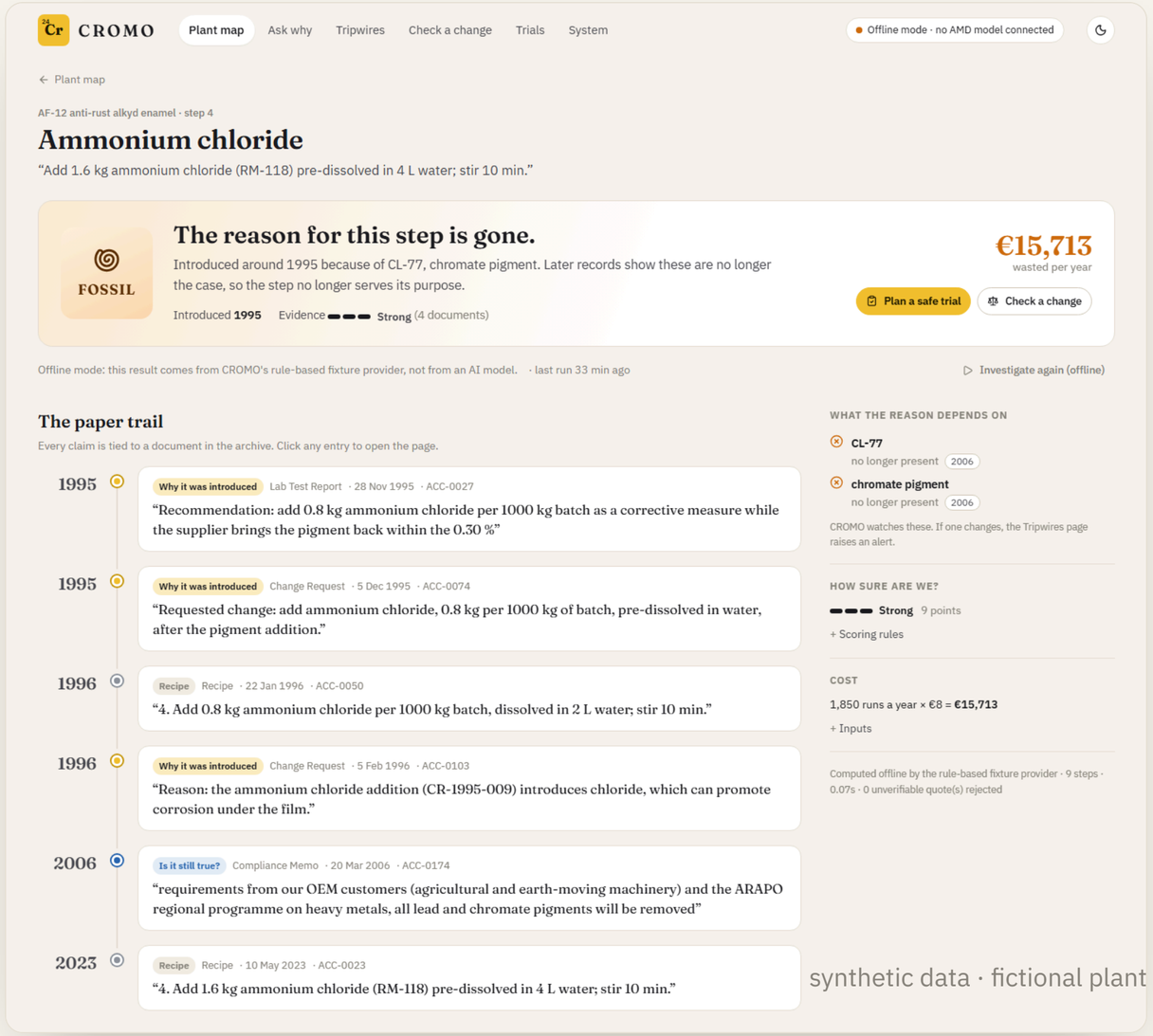
1996 supplier fax: process fixed

2006 memo: chromate pigment phased out

2007 new neutral pigment qualified

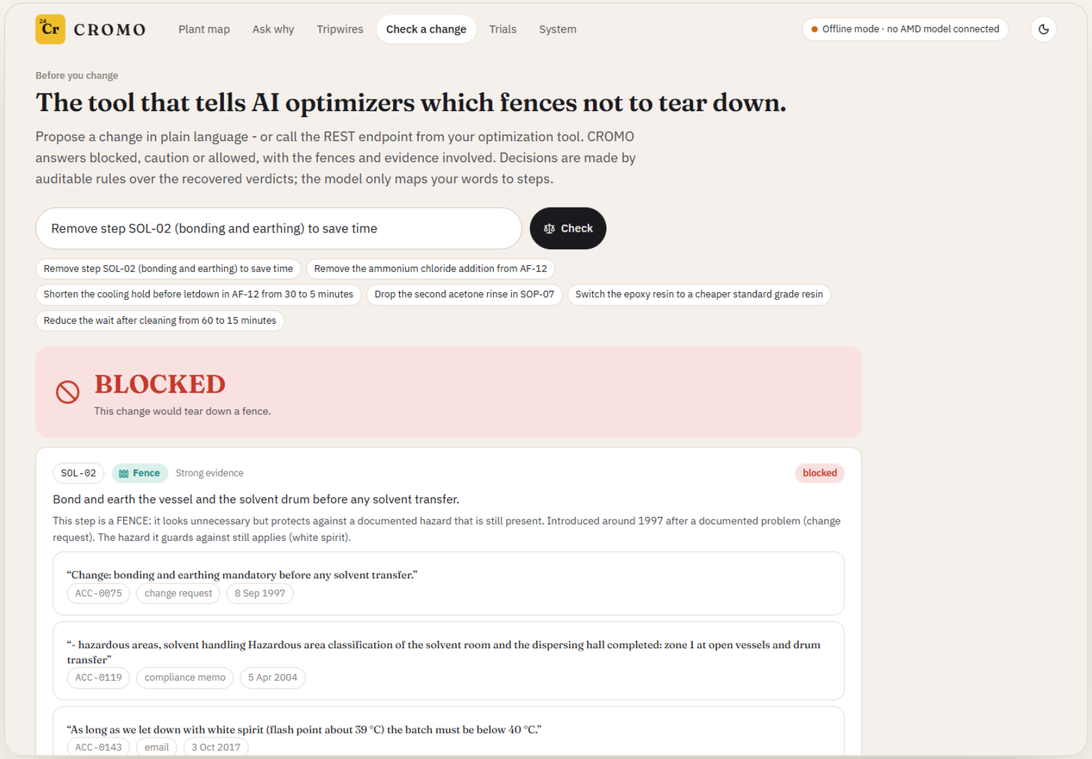
2020 lab: NH4Cl has no effect

FOSSIL · strong evidence · €15,713/yr



synthetic data · fictional plant

Models read the evidence. Rules decide what may be removed.



Every claim cites a quote **verified verbatim** in the source; uncited claims are dropped.

A fossil needs proof that its reason ended and no open safety evidence - otherwise it goes to a person.

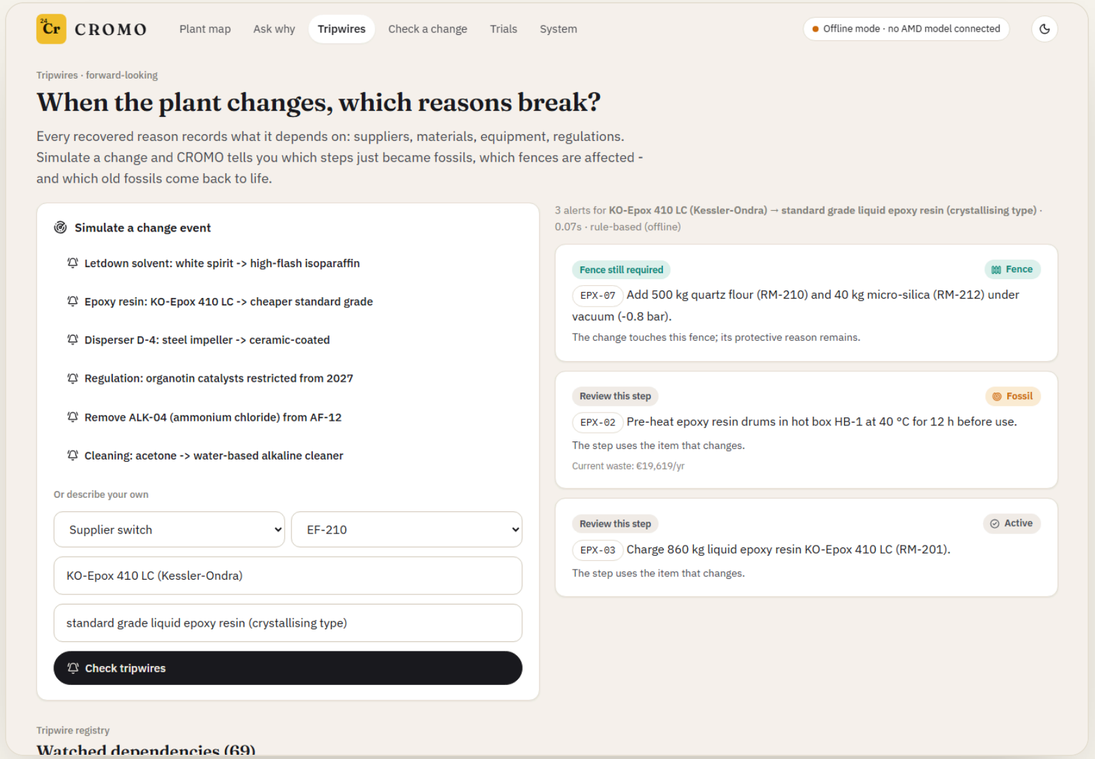
Before you change - also a REST API for optimisers: blocked · caution · allowed with trial.

When the plant changes, which reasons break?

Each recovered reason records its dependencies: suppliers, materials, equipment, regulations.

Switch to a cheaper, crystallising epoxy resin and CROMO warns: the "useless" hot-box step is **needed again**.

Replace a solvent, and the fences that guarded against it may become fossils - re-investigate, trial, approve.



Read the archive. Follow the paper trail. Decide with rules.

Role	Open model	On AMD
Vision: scans, faxes, handwritten Italian logbooks	Gemma 4 26B-A4B-it (Google DeepMind)	vLLM · ROCm · image input
Reasoner: reads passages, follows leads, proposes verdicts	same model	vLLM · ROCm · tool calling
Embeddings: multilingual retrieval	bge-m3	vLLM · ROCm

Gemma 4, Google DeepMind's open model, is both the eyes and the brain: one model reads every scanned, faxed and handwritten page and does all the reasoning, with native image input and tool calling.

Ingestion → hybrid index → multi-hop investigation (9 hops max, streamed) → verified citations → evidence-strength rules → tripwires, change-check, trials.

Measured on AMD · benchmark pending

Real numbers, never estimates.

pending
archive pages / minute ingested

pending
median multi-hop investigation

pending
GPU cost of a full plant audit (49 steps, 178 docs)

Handwriting & scans read by the VLM: character error rate ***pending*** · *pending* model calls per step

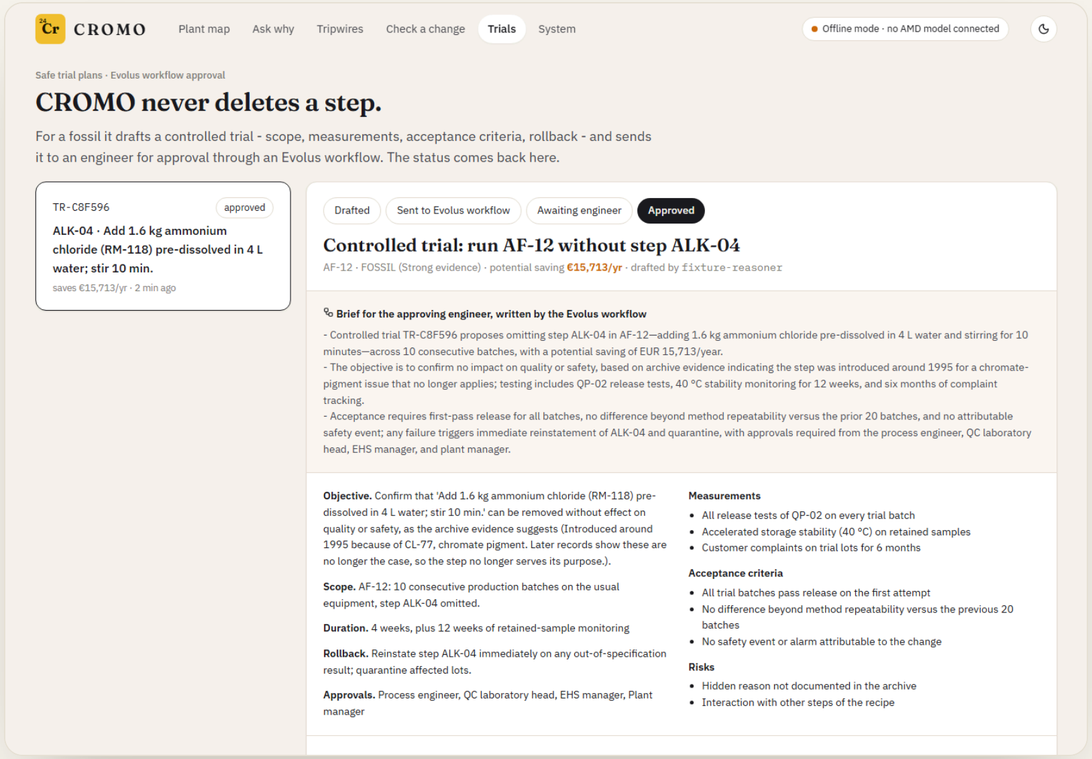
Evaluation · 49 steps with planted ground truth (12 fossils, 9 fences, 22 active, 6 unknown)

Missing a fence is the dangerous error.

	Heuristic baseline (no LLM)	CROMO on AMD
FENCE recall	56%	*pending*
False-FOSSIL rate	11%	*pending*
Accuracy (4 classes)	65%	*pending*
Citation precision	61%	*pending*
Prompt-injection document	ignored	*pending*

Synthetic archive: red herrings, contradictions, a reverted change, a reason only in a handwritten logbook. The product never reads the answer key (tested).

A human approves every change - through Evolus.



Workflow: fossil → trial plan → AI Tool brief for the engineer → approval record → decision → status back in CROMO.

AI Tool compare: what changed when a step was added. **Knowledge library**: the 178-document archive.

All driven through the Evolus REST API; created via POST /workflows.

Find the waste. Keep the fences.

Value: in our synthetic plant, €374,183/yr of fossil steps found, with an audit trail - and every fence protected from well-meant "optimisations".

Model: per-site subscription for process industries (coatings, chemicals, food, pharma); the change-check API for MES and optimisation vendors.

Next: pilot on a real archive · veteran-interview mode for unknowns (voice on AMD) · MES connectors · cross-recipe fossil detection.

Team: solo developer, built with Claude Code. MIT licensed, open models, zero budget.